

Apostol Chapter 1 - Exercise 7

Afonso Vilhena Francisco

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A rational number a/b with $(a, b) = 1$ is called a *reduced fraction*. if the sum of two reduced fractions is an integer, say $(a/b) + (c/d) = n$, prove that $|b| = |d|$.

Suppose that $\frac{a}{b} + \frac{c}{d} = n$, with $(a, b) = 1 = (c, d)$.
Multiplying the previous equation by bd , we get

$$ad + cb = bdn \quad (1)$$

This way we can write both

$$ad = bdn - cb \quad (2)$$

$$cb = bdn - ad \quad (3)$$

Since $(a, b) = 1$, we can write $1 = ax + by$, for some integers x, y . Multiplying it by d and using (2), we get,

$$\begin{aligned} d &= adx + bdy \\ &= (bdn - cb)x + bdy \\ &= (dnx - cx + dy)b, \end{aligned}$$

that is, $b|d$.

A similar argument shows that $d|b$. Indeed, since $(c, d) = 1$, we can write $1 = cz + dw$, for some integers z, w . Again, multiplying it by b and using (3), it follows that,

$$\begin{aligned} b &= cbz + dbw \\ &= (bdn - ad)z + dbw \\ &= (bnz - az + bw)d, \end{aligned}$$

that is, $d|b$.

Finally, since $b|d$ and $d|b$, by item (j) of *Theorem 1.1*, follows that $|b| = |d|$